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Organic acid fractionation of hardwoods planted in social forestry

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Abstract

Trema orientalis and *Trewia nudiflora* are the fastest rotation trees (3–4 years' rotation) in Bangladesh, and experimentally planted in social forestry in Gazipur district. Biorefinery is the most cited topic in research recent era in order to established circular bioeconomy. In established a biorefinery initiative, both species were fractionated into pulp, lignin and hemicellulose by formic acid (FA) treatment at atmospheric conditions. After alkaline peroxide bleaching of FA treated pulp, the yield was 47.0 % and 41.0 % with α -cellulose of 82.39 % and 81.61 %, and residual pentosan content of 9.26 % and 11.13 % for *T. orientalis* and *T. nudiflora*, respectively. The brightness of the pulps was 84.5 % and 85.0 %. Cold alkali extraction of the bleached pulp increased α -cellulose to 92.36 % and 91.29 % which was suitable for cellulose derivatives. Lignin was separated from the spent liquor by adding water. The FA lignin from hardwood replaced 20 % phenol in phenol-formaldehyde resin preparation without sacrifice of shear strength. The lignin separated FA liquor contains mainly pentose sugars, which was subsequently converted to furfural with a yield of 41.39 % and 41.72 % at 170 °C using 0.1 % H₂SO₄ catalyst from *T. orientalis* and *T. nudiflora*, respectively.

Keywords: [hardwood](#); [formic acid treatment](#); [cellulose](#); [resin](#); [furfural](#)

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